

1) A cyber security consulting company has been contracted to perform an initial security assessment of the publicly accessible server scanme.nmap.org. Before conducting any vulnerability assessment, the client requires a detailed enumeration report. Using Nmap, perform host discovery, TCP port scanning, service version detection, operating system fingerprinting, and default script scanning on scanme.nmap.org. Analyze all identified ports and services, determine their business purpose, and evaluate whether they are necessary for public access. Discuss how a threat actor could utilize the information gathered during enumeration to launch further attacks such as service exploitation, brute-force attacks, or vulnerability discovery. Prepare a technical report summarizing your findings, attack surface analysis, risk assessment, and recommendations for securing the server.

➤ Host Discovery:

Command: `nmap -sn scanme.nmap.org`

```
(manasvi@MANASVI)-[~]  
$ nmap -sn scanme.nmap.org  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:37 +0530  
Failed to resolve "scanme.nmap.org".  
WARNING: No targets were specified, so 0 hosts scanned.  
Nmap done: 0 IP addresses (0 hosts up) scanned in 4.63 seconds
```

➤ TCP Port Scanning

Command: `sudo nmap -sS -p- scanme.nmap.org`

```
(manasvi@MANASVI)-[~]  
$ sudo nmap -sS -p- scanme.nmap.org  
[sudo] password for manasvi:  
Sorry, try again.  
[sudo] password for manasvi:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:38 +0530  
Nmap scan report for scanme.nmap.org (45.33.32.156)  
Host is up (0.035s latency).  
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f  
All 65535 scanned ports on scanme.nmap.org (45.33.32.156) are in ignored states.  
Not shown: 47830 filtered tcp ports (net-unreach), 17705 filtered tcp ports (no-response)  
Nmap done: 1 IP address (1 host up) scanned in 74.14 seconds
```

➤ Service Version Detec

Command: `sudo nmap -sV scanme.nmap.org`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sV scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:40 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0083s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 991 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
80/tcp    open  http?
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
8081/tcp  closed blackice-icecap

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 85.52 seconds
```

➤ Operating System Fingerprinting

Command: `sudo nmap -O scanme.nmap.org`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -O scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:41 +0530
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 0 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0054s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
5060/tcp  open  sip
5061/tcp  open  sip-tls
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: switch|phone
Running (JUST GUESSING): Cisco embedded (87%), Nokia Symbian OS (86%)
OS CPE: cpe:/h:cisco:catalyst_1900 cpe:/o:nokia:symbian_os
Aggressive OS guesses: Cisco Catalyst 1900 switch (87%), Nokia 3600i mobile phone (86%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.98 seconds
--(manasvi@MANASVI)-[~]
```

➤ Default Script Scanning

Command: `sudo nmap -sC scanme.nmap.org`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sC scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:42 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0050s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls

Nmap done: 1 IP address (1 host up) scanned in 13.73 seconds
(manasvi@MANASVI)-[~]
```

➤ Recommended Combined Scan

Command: `sudo nmap -sS-Ss -O -Sc scanme.nmap.org`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sS -sV -O -sC scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 10:42 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.0055s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
80/tcp    open  http         Apache httpd 2.4.7
139/tcp   closed netbios-ssn
143/tcp   closed imap
443/tcp   closed https
3389/tcp  closed ms-wbt-server
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
8080/tcp  closed http-proxy
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (90%), Oracle Virtualbox (88%), Slirp (88%), QEMU (86%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (90%), Oracle Virtualbox Slirp NAT bridge (88%), QEMU user mode ne
twork gateway (86%)
No exact OS matches for host (test conditions non-ideal).
Service Info: Host: scanme.nmap.org

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 126.57 seconds
```

2) The security team has deployed a deliberately vulnerable Linux server, 192.168.56.101 (Metasploitable 2), in an isolated laboratory environment to train penetration testers. As part of the pre-engagement reconnaissance phase, use Nmap to perform comprehensive enumeration of 192.168.56.101. Identify all open ports, running services, service versions, and operating system information. Analyze the purpose of each service and determine which services could provide potential entry points for an attacker. Categorize the discovered services into High, Medium, and Low risk levels based on their exposure and functionality. Explain how the enumeration results could guide a penetration tester in selecting potential attack vectors and exploitation strategies.

Host Discovery

Command: `nmap -sn 192.168.56.101`

```
(manasvi@MANASVI)-[~]
$ nmap -sn 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 11:04 +0530
Nmap scan report for 192.168.56.101
Host is up (0.00083s latency).
Nmap done: 1 IP address (1 host up) scanned in 2.17 seconds
```

TCP Port Scan — All Ports

Commands: `sudo nmap -sS -p- 192.168.56.101`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sS -p- 192.168.56.101
[sudo] password for manasvi:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-11 11:05 +0530
Nmap scan report for 192.168.56.101
Host is up (0.079s latency).
Not shown: 44083 filtered tcp ports (net-unreach), 21451 filtered tcp ports (no-response)
PORT      STATE SERVICE
5061/tcp  open  sip-tls
Nmap done: 1 IP address (1 host up) scanned in 89.90 seconds
```

Service & Version Detection

`sudo nmap -sV -p- 192.168.56.101`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sV -p- 192.168.56.101
[sudo] password for manasvi:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 13:35 +0530
Nmap scan report for 192.168.56.101
Host is up (0.10s latency).
All 65535 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 37940 filtered tcp ports (no-response), 27595 filtered tcp ports (net-unreach)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 376.03 seconds
```

Operating System Detection

Commands: `sudo nmap -O 192.168.56.101`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -O 192.168.56.101
[sudo] password for manasvi:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 13:52 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0052s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS010U print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.97 seconds
```

Default Script Scan

Command: `sudo nmap -sC 192.168.56.101`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sC 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 13:57 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0011s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 11.22 seconds
```

Comprehensive Enumeration

Command: `sudo nmap -sS -sV -O -sC 192.168.56.101`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sS -sV -O -sC 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 14:02 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0011s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS010U print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.66 seconds
```

3)A financial organization wants to assess the security posture of a Windows workstation located at 192.168.56.102 before connecting it to the production network. Using Nmap, enumerate 192.168.56.102 to identify active ports, network services, file-sharing protocols, remote management services, and operating system details. Analyze whether the discovered services comply with the principle of least privilege and identify any services that may unnecessarily increase the attack surface. Discuss how an attacker could leverage the identified services for reconnaissance, lateral movement, or unauthorized access. Based on your findings, propose a set of security hardening measures to improve the system's resilience against cyberattacks.

Host Discovery

Command: `nmap -sn 192.168.56.102`

```
(manasvi@MANASVI)-[~]  
$ nmap -sn 192.168.56.102  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 14:16 +0530  
Nmap scan report for 192.168.56.102  
Host is up (0.00085s latency).  
Nmap done: 1 IP address (1 host up) scanned in 4.13 seconds
```

Scan All TCP Ports

Command: `sudo nmap -sS -p- 192.168.56.102`

```
(manasvi@MANASVI)-[~]  
$ sudo nmap -sS -p- 192.168.56.102  
[sudo] password for manasvi:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 14:21 +0530  
Nmap scan report for 192.168.56.102  
Host is up (0.072s latency).  
All 65535 scanned ports on 192.168.56.102 are in ignored states.  
Not shown: 48348 filtered tcp ports (net-unreach), 17187 filtered tcp ports (no-response)  
Nmap done: 1 IP address (1 host up) scanned in 64.21 seconds
```


OS Detection

Command: `sudo nmap -O 192.168.56.102`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -O 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 14:23 +0530
Nmap scan report for 192.168.56.102
Host is up (0.049s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 31.83 seconds
```

Service and Version Detection

Command: `sudo nmap -sV -p- 192.168.56.102`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sV -p- 192.168.56.101
[sudo] password for manasvi:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 13:35 +0530
Nmap scan report for 192.168.56.101
Host is up (0.10s latency).
All 65535 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 37940 filtered tcp ports (no-response), 27595 filtered tcp ports (net-unreach)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 376.03 seconds
```

Default NSE Scripts

Command: `sudo nmap -sC 192.168.56.102`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sC 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 13:57 +0530
Nmap scan report for 192.168.56.101
Host is up (0.0011s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 11.22 seconds
```

Comprehensive Scan

Command: `sudo nmap -sS -sV -O -sC 192.168.56.102`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sS -sV -O -sC 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 14:45 +0530
Nmap scan report for 192.168.56.102
Host is up (0.0022s latency).
All 1000 scanned ports on 192.168.56.102 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AXIS 2100 Network Camera (92%), D-Link DP-300U, DP-G310, or Hamlet HPS01UU print server (92%), HP Tru64 UNIX 5.1A (92%), Sanyo PLC-XU88 digital video projector (92%), TRENDnet TV-IP100 webcam (92%), Vantage HD7100S satellite receiver (92%), Linux 2.6.8 (90%), Asus WL-500gP wireless broadband router (89%), Cisco 1812, 3640, or 3700 router (IOS 12.4) (89%), Cisco DOCSIS cable modem termination server (IOS 12.1) (89%)
No exact OS matches for host (test conditions non-ideal).

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.17 seconds
```

4)An organization hosts a web application on 192.168.56.103 (OWASP Juice Shop Server) and plans to conduct a full penetration test. Before the testing phase begins, the security team must understand the server's network exposure. Using Nmap, perform a detailed enumeration of 192.168.56.103, including host discovery, port scanning, service version detection, operating system fingerprinting, and web service identification. Analyze the discovered services and determine how they support the hosted web application. Evaluate the risks associated with exposed web services, administrative interfaces, and supporting services. Explain how the information gathered during enumeration could be used to identify potential vulnerabilities in the web application environment and recommend measures to minimize exposure

Host Discovery

Command: `nmap -sn 192.168.56.103`

```
(manasvi@MANASVI)-[~]
$ nmap -sn 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 15:22 +0530
Nmap scan report for 192.168.56.103
Host is up (0.00072s latency).
Nmap done: 1 IP address (1 host up) scanned in 4.16 seconds
```


Scan All TCP Ports

Command: `sudo nmap -sS -p- 192.168.56.103`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sS -p- 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 17:02 +0530
Nmap scan report for 192.168.56.103
Host is up (0.083s latency).
Not shown: 50543 filtered tcp ports (net-unreach), 14991 filtered tcp ports (no-response)
PORT      STATE SERVICE
5061/tcp  open  sip-tls
Nmap done: 1 IP address (1 host up) scanned in 54.48 seconds
```

Service Version Detection

Command: `sudo nmap -sV -p- 192.168.56.103`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sV -p- 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 17:04 +0530
Nmap scan report for 192.168.56.103
Host is up (0.051s latency).
Not shown: 57769 filtered tcp ports (net-unreach), 7765 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
5060/tcp  open  sip?
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 124.36 seconds
```

Operating System Detection

Command: `sudo nmap -O 192.168.56.103`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -O 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 17:07 +0530
Nmap scan report for 192.168.56.103
Host is up (0.0057s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.29 seconds
```

Default Script Scan

Command: `sudo nmap -sC 192.168.56.103`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sC 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 17:08 +0530
Nmap scan report for 192.168.56.103 scanned in 4.16 seconds
Host is up (0.0029s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
5060/tcp  open  sip
5061/tcp  open  sip-tls
Nmap done: 1 IP address (1 host up) scanned in 71.40 seconds
```

Web Service Identification

Command: `sudo nmap -p 80,443,3000,8080,8000 -sV 192.168.56.103`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -p 80,443,3000,8080,8000 -sV 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 17:11 +0530
Nmap scan report for 192.168.56.103
Host is up (0.00075s latency).
PORT      STATE SERVICE VERSION
80/tcp    filtered http
443/tcp    filtered https
3000/tcp   filtered ppp
8000/tcp   filtered http-alt
8080/tcp   filtered http-proxy
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.64 seconds
```

Comprehensive Enumeration

Command: `sudo nmap -sS -sV -O -sC 192.168.56.103`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sS -sV -O -sC 192.168.56.103
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 17:12 +0530
Nmap scan report for 192.168.56.103
Host is up (0.0053s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
5060/tcp  open  sip?
5061/tcp  open  sip-tls?
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 156.46 seconds
```

5)The network gateway device at 192.168.56.1 serves as the primary access point between the internal laboratory network and external networks. The network administrator suspects that unnecessary management services may be exposed. Using Nmap, conduct a comprehensive enumeration of 192.168.56.1, including port scanning, service detection, operating system,fingerprinting, and script-based enumeration. Identify all exposed management interfaces, network services, and accessible resources. Analyze the security implications of exposing services such as SSH, Telnet, SNMP, HTTP, or HTTPS management interfaces. Discuss how attackers could exploit misconfigured management services to gain unauthorized access to the network infrastructure. Based on your analysis, recommend security controls and best practices to secure the gateway device and reduce the risk of compromise.for this question

Host Discovery

Command: nmap -sn 192.168.56.1

```
(manasvi@MANASVI)-[~]  
$ nmap -sn 192.168.56.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 16:19 +0530  
Nmap scan report for 192.168.56.1  
Host is up (0.00023s latency).  
Nmap done: 1 IP address (1 host up) scanned in 0.64 seconds
```

Scan All TCP Ports

Command: `sudo nmap -sS -p- 192.168.56.1`

```
(manasvi@MANASVI)-[~] 192.168.56.102
$ sudo nmap -sS 192.168.56.1map.org ) at 2026-08-12 14:43 +0530
[sudo] password for manasvi:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 16:15 +0530
Nmap scan report for 192.168.56.1
Host is up (0.0014s latency).192.168.56.102
Not shown: 996 filtered tcp ports (no-response)-08-12 14:45 +0530
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds 192.168.56.102
1025/tcp   open  NFS-or-IIS-:/nmap.org ) at 2026-08-12 14:45 +0530
Nmap scan report for 192.168.56.102
Nmap done: 1 IP address (1 host up) scanned in 6.06 seconds
```

Service and Version Detection

Command: `sudo nmap -sV -p- 192.168.56.1`

```
(manasvi@MANASVI)-[~] 192.168.56.102
$ sudo nmap -Pn -sV 192.168.56.1 conditions non-ideal).
[sudo] password for manasvi:
Sorry, try again. action performed. Please report any incorrect results at https://nmap.org/submit/ .
[sudo] password for manasvi:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 16:44 +0530
Nmap scan report for 192.168.56.1
Host is up (0.0023s latency).
Not shown: 996 filtered tcp ports (no-response) 08-12 15:32 +0530
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
1025/tcp   open  msrpc        Microsoft Windows RPC
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 15.14 seconds
```

Operating System Fingerprinting

Command: `sudo nmap -O 192.168.56.1`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -O 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 16:46 +0530
Nmap scan report for 192.168.56.1
Host is up (0.0016s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
1025/tcp   open  NFS-or-IIS
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.15 seconds
```

Default Script Enumeration

Command: `sudo nmap -sC 192.168.56.1`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sC 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 16:49 +0530
Nmap scan report for 192.168.56.1
Host is up (0.0017s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
1025/tcp   open  NFS-or-IIS
Host script results:
| smb2-time: IP address (1 host up) scanned in 4.16 seconds
|   date: 2026-08-12T11:19:50
|_  start_date: N/A
| smb2-security-mode: 192.168.56.1
|_  3.1.1: Message signing enabled and required
Nmap done: 1 IP address (1 host up) scanned in 39.77 seconds
```

UDP Port Scan

Command: `sudo nmap -sU --top-ports 100 192.168.56.1`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sU --top-ports 100 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 16:55 +0530
Nmap scan report for 192.168.56.1 (org ) at 2026-08-12 15:22 +0530
Host is up (0.00093s latency).
Not shown: 96 filtered udp ports (port-unreach)
PORT      STATE SERVICE
138/udp    open|filtered netbios-dgm
1900/udp   open|filtered upnp
4500/udp   open|filtered nat-t-ike
5353/udp   open|filtered zeroconf
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 15:42 +0530
Nmap done: 1 IP address (1 host up) scanned in 9.01 seconds
```

Management-Service Scan

Command: `sudo nmap -sV -p 22,23,80,161,443 192.168.56.1`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sV -p 22,23,80,161,443 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 16:59 +0530
Nmap scan report for 192.168.56.1
Host is up (0.00068s latency).
Nmap scan report for 192.168.56.103
PORT      STATE SERVICE VERSION
22/tcp    filtered ssh
23/tcp    filtered telnet
80/tcp    filtered http
161/tcp   filtered snmp
443/tcp   filtered https
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 15:42 +0530
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.41 seconds
(manasvi@MANASVI)-[~]
```


Management-Service NSE Enumeration

Command: `sudo nmap -sV -p 22,23,80,161,443 --script banner,http-title,http-headers,snmp-info 192.168.56.1`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sV -p 22,23,80,161,443 --script banner,http-title,http-headers,snmp-info 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 17:00 +0530
Nmap scan report for 192.168.56.1
Host is up (0.00075s latency).
Nmap scan report for 192.168.56.1
PORT      STATE SERVICE VERSION
22/tcp    filtered ssh
23/tcp    filtered telnet
80/tcp    filtered http
161/tcp    filtered snmp
443/tcp    filtered https
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 5.52 seconds
```

Comprehensive Enumeration

Command: `sudo nmap -sS -sV -O -sC 192.168.56.1`

```
(manasvi@MANASVI)-[~]
$ sudo nmap -sS -sV -O -sC 192.168.56.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-12 17:01 +0530
Nmap scan report for 192.168.56.1
Host is up (0.00090s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
135/tcp    open  msrpc      Microsoft Windows RPC
139/tcp    open  netbios-ssn Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
1025/tcp   open  msrpc      Microsoft Windows RPC
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose
Running (JUST GUESSING): AT&T embedded (93%), Oracle Virtualbox (92%), Slirp (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (93%), Oracle Virtualbox Slirp NAT bridge (92%), QEMU user mode network gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Service Info: OS: Windows; CPE: o:microsoft:windows

Host script results:
|_ smb2-security-mode:
|_ 3.1.1:
|_ Message signing enabled and required
|_ smb2-time:
|_ date: 2026-08-12T11:31:56
|_ start_date: N/A

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 26.99 seconds
```